

RATANDEEP REDDY ALLGUPALLY

(602) 577-2602 • ratandeep569@gmail.com • Tempe, AZ

Professional Summary

Mechanical Engineering student at Arizona State University with hands-on experience in CAD design, structural analysis, prototyping, and engineering simulation. Skilled in SolidWorks, ANSYS FEA, MATLAB, and embedded systems. Experience developing mechanical prototypes and analyzing system performance through experimental testing and simulation. Seeking an engineering opportunity to contribute to high-performance vehicle systems and advanced mechanical design in motorsport environments.

EDUCATION

Arizona State University, Tempe, Arizona

Aug 2023 - May 2027

Bachelor of Science in Engineering (B.S.E)

(GPA 3.75/4.00)

Relevant Coursework: Thermodynamics • Fluid Mechanics • Mechanics of Materials • Manufacturing Processes • Mechatronics • Dynamic Systems & Controls • Signals & Systems • Engineering Statistics

TECHNICAL SKILLS

- **Engineering & Analysis Tools:** SolidWorks, AutoCAD, ANSYS (FEA), MATLAB, Excel
- **Programming & Systems:** Python, MATLAB, C/C++, Arduino (Embedded C)
- **Engineering Methods:** Root cause analysis, quality inspection, DFM, tolerance analysis, preventative maintenance, numerical integration
- **Fabricating & Testing:** 3D printing (FDM), load testing, stress validation, dimensional inspection, prototype development

EXPERIENCE

Mechanical Engineering Intern

Hyderabad, India

Sreenex Machine Private Limited

May 2025 – August 2025

- Designed and optimized 3D CAD models, reducing design-revision cycles by 25% through improved tolerance control
- Built and evaluated 15+ mechanical prototypes; identified structural failure points and improved reliability by 30%
- Conducted load testing, stress validation, and dimensional inspections, reducing defect rates by 18%
- Supported preventative maintenance checks and safety compliance in fabrication operations
- Created GD&T drawings and BOMs to standardize workflows and accelerate approvals by 20%

Undergraduate Research Assistant

Tempe, Arizona

Arizona State University

August 2025 – present

- Designed and built functional prototypes for experimental validation of mechanical systems
- Developed experimental setups and integrated instrumentation for mechanical testing and structured data acquisition
- Collected and analyzed performance data to evaluate system behavior, identify inefficiencies, and improve reliability
- Improved manufacturing workflow efficiency by 20% through bottleneck identification and process restructuring
- Conducted quality inspections on 200+ components, implemented DFM improvements, reducing waste by 12% and minimizing rework

Technology Consultant

Tempe, Arizona

Arizona State University

November 2024 – August 2025

- Diagnosed and resolved hardware and software issues in a fast-paced technical support environment
- Managed and tracked technology inventory, conducting regular audits to ensure accurate stock levels and equipment availability
- Assisted with CAD modeling and engineering simulations using SolidWorks, AutoCAD, and ANSYS
- Supported projects involving process optimization, digital manufacturing, and the integration of emerging technologies like IoT

PROJECTS

Aerodynamic Active RC Car (In-progress)

- Designing a high-performance RC car powered by a brushless drone motor; modeling chassis and body in SolidWorks, programming control logic via Arduino, and simulating aerodynamic behavior in ANSYS to optimize downforce and reduce drag

Dynamic Objects Using Seal-Whisker Shaped Sensors (Volunteer Research Assistant) – Arizona State University

- Designed and built a Doppler-effect setup, recorded acoustic data, and analyzed frequency patterns using Logger Pro and Excel
- Gained experience in experimental design, data processing, and signal analysis, inspiring further research through the FURI program

Water Tank Outlet Radius Simulation – MAE 384 Project | Spring 2025

- Developed a MATLAB simulation of a gravity-dependent tank system, integrating DFT-modeled $g(t)$, RK4 ODE solving, spline interpolation, and Simpson's rule, and calibrated the optimal outlet radius using convergence testing and bisection root-solving

Solar Flipping Mechanism Using Weight and Gears – MAE 202 Project | Spring 2025

- Engineered a non-electrical, weight-powered solar panel flipping mechanism by applying torque and rotational dynamics, optimizing gear design through iterative prototyping to achieve smooth and energy-efficient rotation

Automated Irrigation System | Summer 2025

- Designed an Arduino-based automated irrigation system using soil moisture sensors and servo-controlled valves to regulate water flow
- Model the enclosure in SolidWorks for component integration and conducted reliability testing to ensure consistent long-term performance